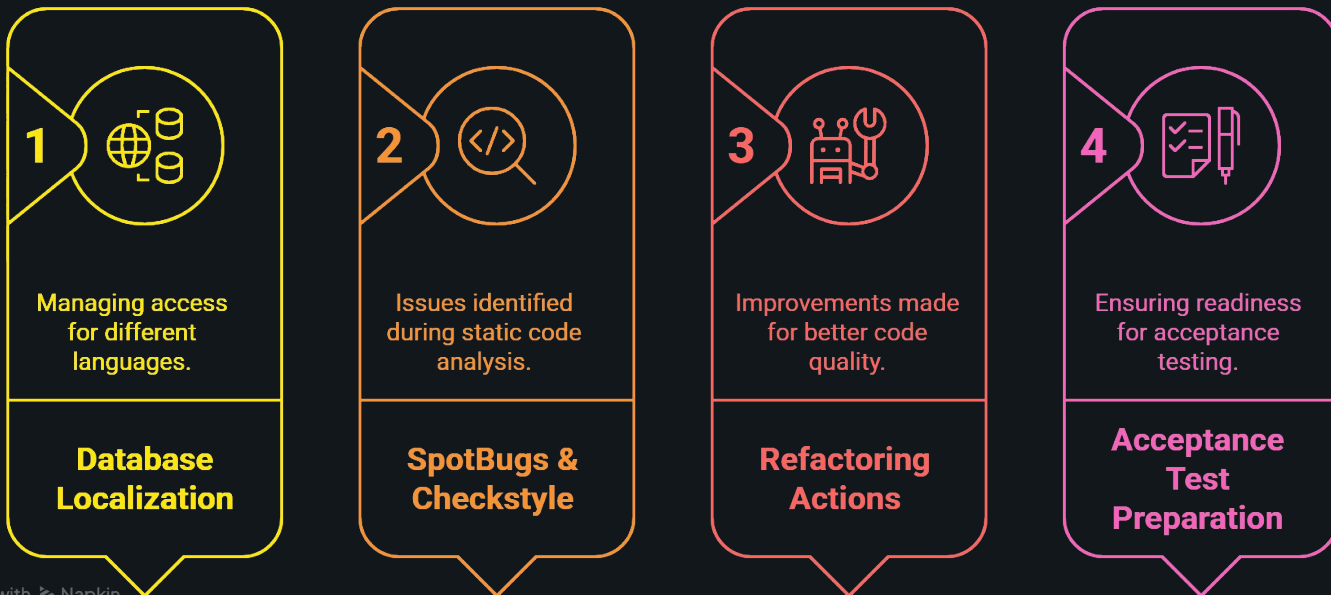




Scrum Review for Sprint 6

StockMaster3000

Key Implementations



DB Localisation

language_code
en
ru
en
en

// Get all products in a specific inventory

```
public List<Product> getProductsByInventory(Long inventoryId) {  
    return productRepository.findByInventoryId(inventoryId);  
}
```

// Fetching Product by Name

```
public List<Product> getProductsByName(Long inventoryId, String productName) {  
    if (productName != null && !productName.isEmpty()) {  
        return productRepository.findByInventoryIdAndName(inventoryId, productName);  
    } else {  
        return productRepository.findByInventoryId(inventoryId);  
    }  
}
```

// Get all products in a specific inventory

```
public List<Product> getProductsByInventory(Long inventoryId, String languageCode) {  
    return productRepository.findByInventoryIdAndLanguageCode(inventoryId, languageCode);  
}
```

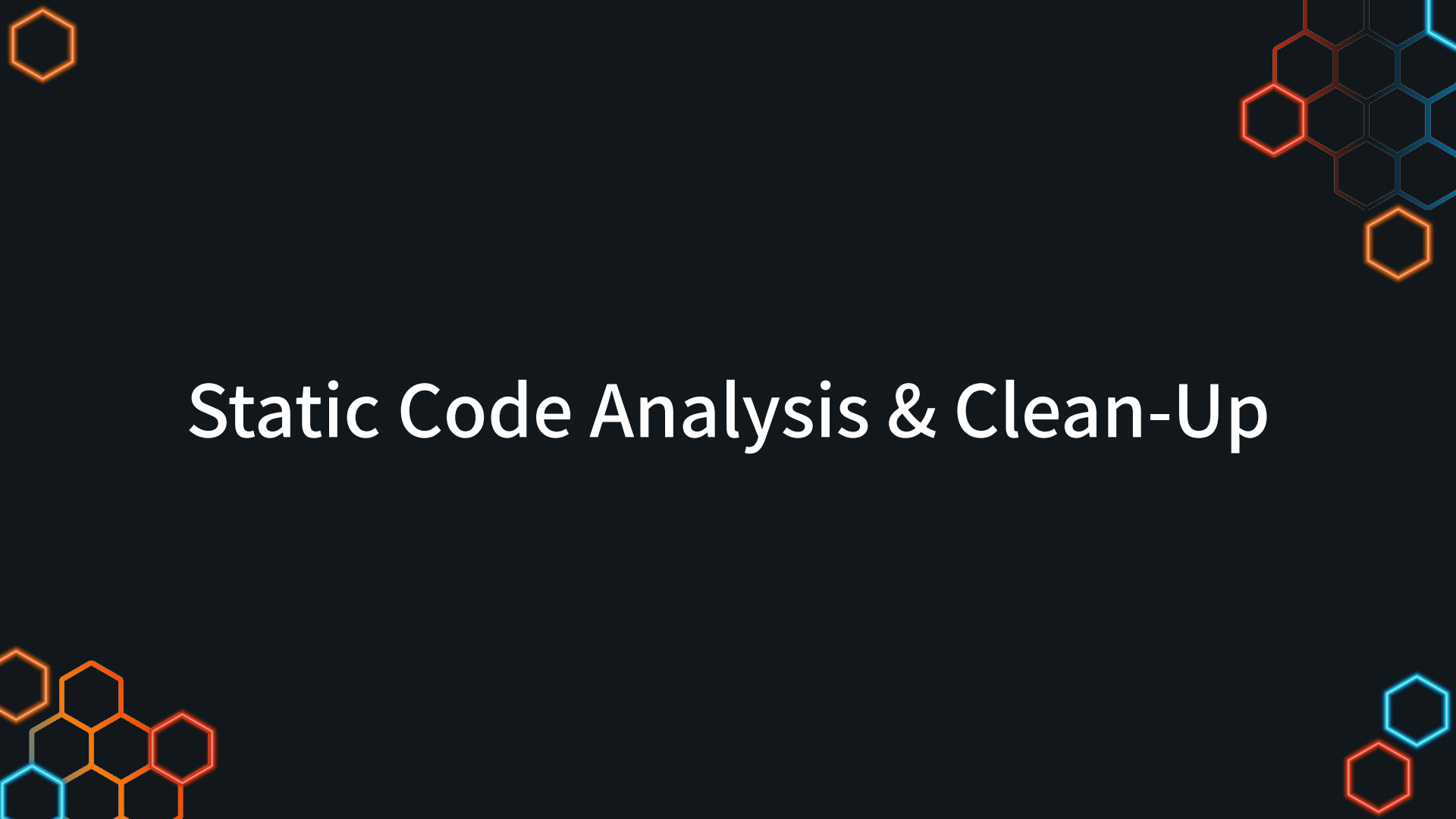
// Fetching Product by Name

```
public List<Product> getProductsByName(Long inventoryId, String productName, String languageCode) {  
    if (productName != null && !productName.isEmpty()) {  
        return productRepository.findByInventoryIdAndName(inventoryId, productName);  
    } else {  
        return productRepository.findByInventoryIdAndLanguageCode(inventoryId, languageCode);  
    }  
}
```

// Set up inventory selection listener

```
inventorySelectorComponent.setSelectionListener(selectedInventory -> {  
    inventoryComponent.updateGrid(selectedInventory, currentLanguageCode); // Update In  
    inventoryChartComponent.updateCharts(selectedInventory); // Update ALL charts  
});
```

products
columns 12
id bigint (auto increment)
name varchar(255)
price double
quantity int
nutritions text
amountOfDaysUntilExpiration int
language_code varchar(5) = 'en'
supplier_id bigint
category_id bigint
inventory_id bigint
created_at timestamp = current_timestamp()
updated_at timestamp = current_timestamp()

The slide features a dark blue background with decorative hexagonal patterns in the corners. The top-left corner has a single orange hexagon. The top-right corner has a cluster of orange and blue hexagons. The bottom-left corner has a cluster of orange and blue hexagons. The bottom-right corner has a small cluster of orange and blue hexagons.

Static Code Analysis & Clean-Up

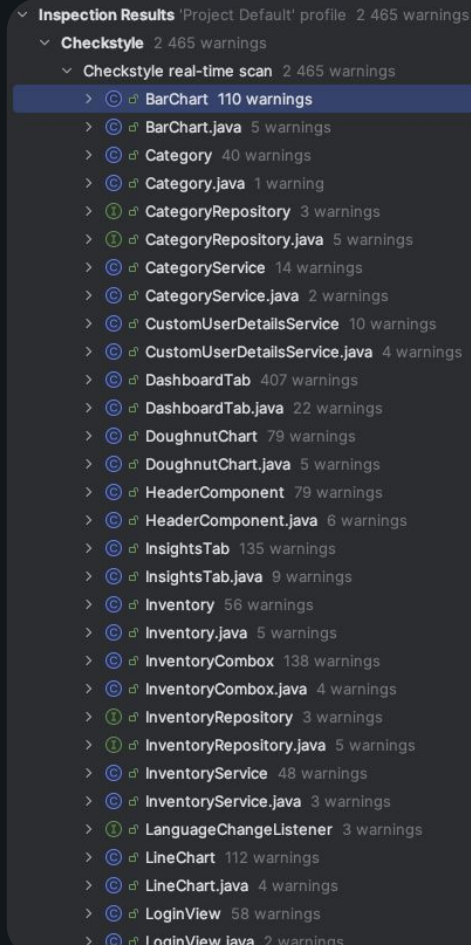
CheckStyle Analysis

Auto/Semi-Auto Fixable Issues (~85% resolved automatically):

- **Indentation (~1500-2000 issues):**
All files used 4-space indentation, while the Google style requires **2-space indentation**.
- **Import Order (~150-200 issues):**
Imports were not in the required order (e.g., static imports, java.* before third-party libs).

Manually Reviewed / Partially Resolved Issues (~15%):

- **Operator Placement (~100 issues):**
“++” or “&&” had to be at the start of the line.
- **Line Length (>100 characters) (~100-200 issues):**
Long lines were found in method chains, log statements, or embedded SQL.
- **Javadoc Missing/Invalid (~100 issues):**
Many public methods and classes lacked Javadoc or had incomplete ones.



```
Inspection Results 'Project Default' profile: 2 465 warnings
  Checkstyle 2 465 warnings
    Checkstyle real-time scan 2 465 warnings
      > [C] BarChart 110 warnings
      > [C] BarChart.java 5 warnings
      > [C] Category 40 warnings
      > [C] Category.java 1 warning
      > [I] CategoryRepository 3 warnings
      > [I] CategoryRepository.java 5 warnings
      > [C] CategoryService 14 warnings
      > [C] CategoryService.java 2 warnings
      > [C] CustomUserDetailsService 10 warnings
      > [C] CustomUserDetailsService.java 4 warnings
      > [C] DashboardTab 407 warnings
      > [C] DashboardTab.java 22 warnings
      > [C] DoughnutChart 79 warnings
      > [C] DoughnutChart.java 5 warnings
      > [C] HeaderComponent 79 warnings
      > [C] HeaderComponent.java 6 warnings
      > [C] InsightsTab 135 warnings
      > [C] InsightsTab.java 9 warnings
      > [C] Inventory 56 warnings
      > [C] Inventory.java 5 warnings
      > [C] InventoryCombox 138 warnings
      > [C] InventoryCombox.java 4 warnings
      > [I] InventoryRepository 3 warnings
      > [I] InventoryRepository.java 5 warnings
      > [C] InventoryService 48 warnings
      > [C] InventoryService.java 3 warnings
      > [I] LanguageChangeListener 3 warnings
      > [C] LineChart 112 warnings
      > [C] LineChart.java 4 warnings
      > [C] LoginView 58 warnings
      > [C] LoginView.java 2 warnings
```

Examples

Rules: Google Checks

Checkstyle found 365 item(s) in 38 file(s)

BarChart.java : 4 item(s)

- ⚠ Extra separation in import group before 'java.util.Arrays' (9:1) [CustomImportOrder] <Google Checks>
- ⚠ Missing a Javadoc comment. (38:3) [MissingJavadocMethod] <Google Checks>
- ⚠ Missing a Javadoc comment. (133:3) [MissingJavadocMethod] <Google Checks>
- ⚠ Line is longer than 100 characters (found 122). (134:0) [LineLength] <Google Checks>

Category.java : 7 item(s)

- ⚠ Using the '*.*' form of import should be avoided - jakarta.persistence.*. (3:27) [AvoidStarImport] <Google Checks>
- ⚠ Extra separation in import group before 'java.util.ArrayList' (5:1) [CustomImportOrder] <Google Checks>
- ⚠ Missing a Javadoc comment. (9:1) [MissingJavadocType] <Google Checks>
- ⚠ Missing a Javadoc comment. (23:3) [MissingJavadocMethod] <Google Checks>

Checkstyle found 12 item(s) in 4 file(s)

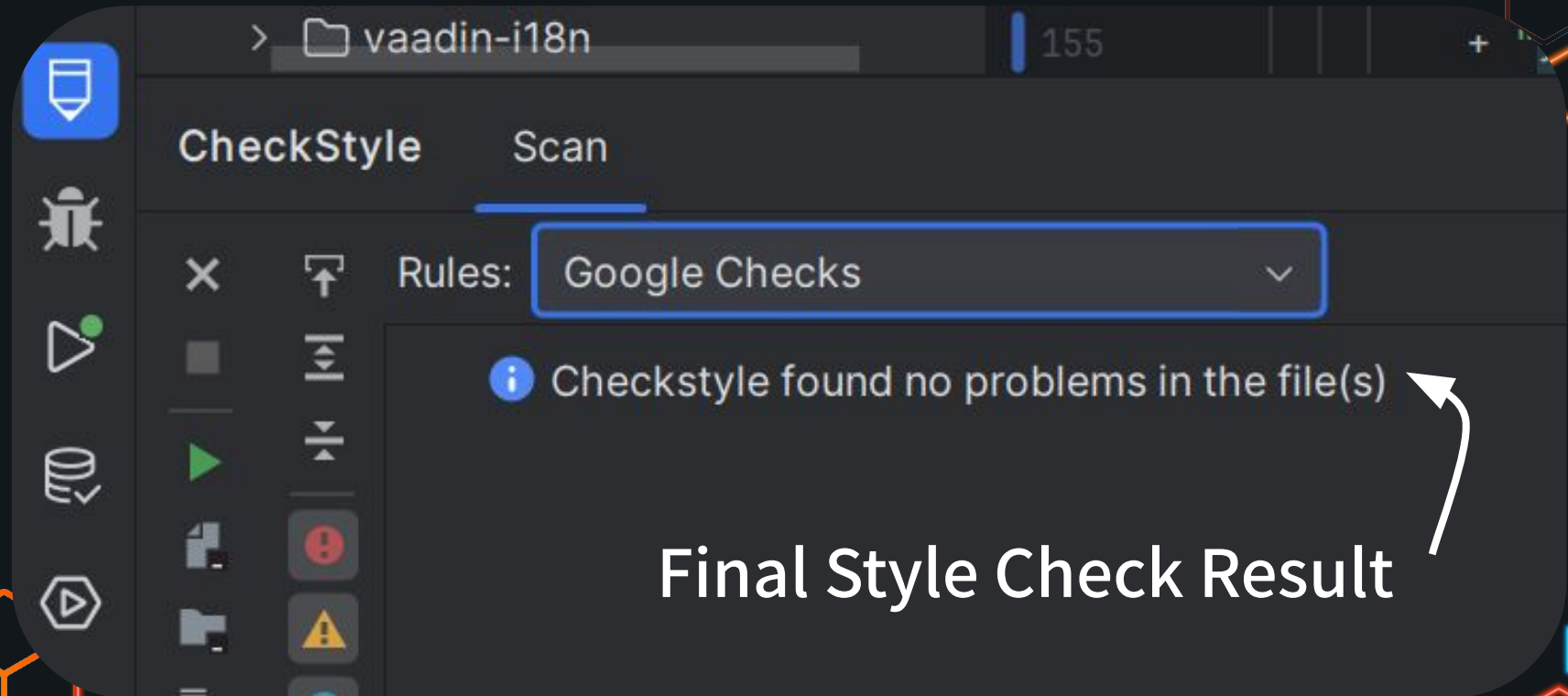
LineChart.java : 4 item(s)

- ⚠ Abbreviation in name 'YAxel' must conform to the Javadoc standard
- ⚠ Local variable name 'YAxel' must match the class name

(12:1) [MissingJavadocType] <Google Checks>
(15:3) [MissingJavadocMethod] <Google Checks>
(17:3) [MissingJavadocMethod] <Google Checks>
(19:3) [MissingJavadocMethod] <Google Checks>
(21:3) [MissingJavadocMethod] <Google Checks>
(23:3) [MissingJavadocMethod] <Google Checks>
(26:3) [MissingJavadocMethod] <Google Checks>

⚠ Line is longer than 100 characters (found 159). (102:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 122). (130:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 104). (179:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 116). (180:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 107). (181:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 122). (182:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 161). (183:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 138). (184:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 138). (185:0) [LineLength] <Google Checks>
⚠ Line is longer than 100 characters (found 115). (195:0) [LineLength] <Google Checks>

Clean-Up Result



The background is a dark navy blue. In the corners, there are decorative patterns of hexagons. Top-left: a single orange-outlined hexagon. Top-right: a cluster of hexagons, some orange and some light blue, connected by lines. Bottom-left: a cluster of orange and light blue outlined hexagons. Bottom-right: two orange-outlined hexagons and one light blue-outlined hexagon.

Security & Code Quality Improvements

Examples

- ✓ StockMaster3000Saed (found 44 bug items in 50 classes)
 - > Malicious code vulnerability (37 items)
 - ✓ Dodgy code (1 item)
 - ✓ Dubious method invocation (1 item)
 - ✓ Code contains a hard coded reference to an absolute
- ✓ Performance (6 items)

© Hard coded reference to an absolute pathname

Preview OpenAiClient.java:

```
55 String absPath = "C:/Users/Käyttaja/Desktop/StockMaster3000/StockMaster3000";
56  if (Files.exists(Paths.get(absPath))) {
57     dotenv = Dotenv.configure()
58         .directory( path: "C:/Users/Käyttaja/Desktop/StockMaster3000/StockMaster3000"
59         .ignoreIfMissing()
60         .load();
61
62     return dotenv;
63 }
64
```

> Malicious code vulnerability (37 items)

> Dodgy code (1 item)

✓ Performance (3 items)

✓ Unused field (1 item)

✓ Unused field (1 item)

© Unused field: org.stockmaster3000.stockmaster3000

✓ Unread field (2 items)

✓ Unread field (2 items)

© Unread field: org.stockmaster3000.stockmaster3000.tab.DashboardTab

© Unread field: org.stockmaster3000.stockmaster3000

Unread field: org.stockmaster3000.stockmaster3000

main > java > org > stockmaster3000 > stockmaster3000 > tab

Preview DashboardTab.java:

```
105 CategoryService categoryService;
106 this.headerComponent = headerComponent;
107 this.securityService = securityService;
108  this.inventoryService = inventoryService;
109 this.productService = productService;
110 this.categoryService = categoryService;
111 this.supplierService = supplierService;
112 this.inventoryComponent = new InventoryComponent();
```

> Malicious code vulnerability (34 items)

> Method returning array may expose internal representation

✓ Storing reference to mutable object (22 items)

✓ May expose internal representation by incorporating reference

© org.stockmaster3000.stockmaster3000.model.Category

© org.stockmaster3000.stockmaster3000.model.Inventory

© new org.stockmaster3000.stockmaster3000.model.Inventory

© org.stockmaster3000.stockmaster3000.model.Inventory

© org.stockmaster3000.stockmaster3000.model.Product

© new org.stockmaster3000.stockmaster3000.model.Product

© org.stockmaster3000.stockmaster3000.model.Product

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© new org.stockmaster3000.stockmaster3000.model.Product

© org.stockmaster3000.stockmaster3000.model.Product

© new org.stockmaster3000.stockmaster3000.model.Product

This code constructs a File object using a

SpotBugs on current state

Category	Count	Severity	Notes
Exposing Internal Representation (Getters)	8	Medium	Methods returning references to mutable objects (e.g. <code>getSupplier()</code>)
Storing Mutable Objects (Setters/Fields)	17	Medium	Assigning external mutable objects directly (e.g. <code>setUser()</code>)
Total Bugs Found	25	Medium	Spotted across 40 classes

```
✓ 🔍 StockMaster3000 (found 25 bug items in 40 classes) more...  
  ✓ [🔴] Malicious code vulnerability (25 items)  
    > 📁 Method returning array may expose internal representation (8 items)  
    > 📁 Storing reference to mutable object (17 items)
```

SonarQube was challenging to get started with due to its many configuration nuances and unfamiliar interface

main 2.8k Lines of Code · Version 1.0 · [Set as homepage](#) [Take the Tour](#)

Quality Gate ⓘ **Failed** Last analysis 3 minutes ago

⚠ The last analysis has warnings. [See details](#)

New Code 1 failed Overall Code

Security 0 Open issues A	Reliability 0 Open issues A	Maintainability 59 Open issues
Accepted issues 0 ⓘ Valid issues that were not fixed	Coverage ⚪ Not computed	Duplications 5.9% On 3.7k lines.

```
pom.xml 1, M sonar-project.properties M InsightsTab.java M
sonar-project.properties
1 sonar.projectKey=myproject
2 sonar.projectName=Stockmaster3000
3 sonar.projectVersion=1.0
4
5 sonar.sources=src/main/java
6 sonar.tests=src/test/java
7 sonar.java.binaries=target/classes
8 sonar.coverage.exclusions=**/.idea/**, **/.mvn/**, **/components/**
9
10
11 sonar.host.url=http://localhost:9000
12 sonar.login=sqp_a9b6102fbffa8d29fac313f4edda4a5979528ec5
13
```

main 2.8k Lines of Code · Version 1.0 · [Set as homepage](#) [Take the Tour](#)

Quality Gate ⓘ **Passed** Last analysis 30 seconds ago

⚠ The last analysis has warnings. [See details](#)

New Code Overall Code

New Code: Since April 10, 2025 Started 1 day ago

New issues 0 Required = 0	Accepted issues 0 Valid issues that were not fixed	
Coverage Not computed	Duplications 0.0% Required ≤ 3.0% On 211 New Lines.	Security Hotspots 0 A

```
// Chart Containers
```

```
doughnutChart = new DoughnutChart(new String[]{getTranslation("insights.noData")}, new int[]{1});
```

Define a constant instead of duplicating this literal "insights.noData" 10 times.

```
spendingChart = new BarChart(new String[]{getTranslation("insights.noData")}, new double[]{0.0});
```

```
expirationChart = new LineChart(new String[]{getTranslation("insights.noData")}, new int[]{0});
```

Acceptance Test Plan – StockMaster3000

1. Introduction

This Acceptance Test Plan outlines the criteria and tests to validate the StockMaster3000 inventory management system. The goal is to ensure the system meets the specified requirements before final deployment.

2. Acceptance Criteria

ID	Requirement
AC1	Users can add, view, update, and delete products.
AC2	Products must have valid supplier, category, and inventory IDs.
AC3	Nutritional info and expiration dates must be saved and displayed correctly.
AC4	The user interface must be intuitive and easy to navigate.
AC5	The system must respond to user actions (e.g., form submissions) within 2 secs.
AC6	The system must prevent invalid data (e.g., negative quantities or empty names).
AC7	The system must store data securely in the database.

3. Acceptance Tests

Functional Tests

Test Case ID	Description	Steps	Expected Result
TC001	Add new product	Fill out product form → Submit	Product is saved and appears in product list
TC002	View product list	Go to product list page	All saved products are listed
TC003	Delete a product	Click delete on an existing product	Product is removed from the database and UI
TC004	Product must have valid supplier/category	User defines the product Category	Product saved with correct foreign keys
TC005	Prevent invalid product input	Leave name blank or enter negative quantity → Submit	Error message is shown, product is not saved

Acceptance test planning

Usability Tests

Test Case ID	Description	Steps	Expected Result
TC006	UI navigation is intuitive	Try adding, viewing, deleting a product without training	User can use the system easily
TC007	Form fields are labeled clearly	View product form	Fields are understandable without confusion

Performance Tests

Test Case ID	Description	Steps	Expected Result
TC008	Form submission time	Submit product form	Response occurs in less than 4 seconds
TC009	List loading time	Open product list with many entries	Page loads in under 2 seconds

4. Test Environment

- OS: macOS (Apple M3 Pro)
- IDE: IntelliJ IDEA, Visual Studio Code
- Browser: Chrome / Safari
- DB: MariaDB
- Language: Java, Vaadin
- Tools: Git, SonarQube, Check Style, FindBugs

5. Sign-off Criteria

The product will be considered accepted if:

- All critical and high-priority test cases pass.
- There are no critical bugs or issues that prevent usage.
- The UI is usable without developer assistance.
- Stakeholders (instructor/team) approve the results.

Jira Backlog Status

- User Stories: 3.5 out of 4
- User Story Points: 11 out of 14

☐	▼	SCRUM-35	Database Localization	DONE	SCRUM Sprint 6	
☐		SCRUM-39	Create new tables for languages	DONE	SCRUM Sprint 6	VT Viet Tran
☐		SCRUM-40	Define backend for new tables	DONE	SCRUM Sprint 6	SA Saed Abukar
☐		SCRUM-41	Predefine user categories	DONE	SCRUM Sprint 6	VT Viet Tran
☐	▼	SCRUM-33	Code Maintenance	IN PROGRESS		
☐	▼	SCRUM-36	Statistical Code Review	DONE	SCRUM Sprint 6	
☐		SCRUM-43	Run Static Code Analysis Tools	DONE	SCRUM Sprint 6	Pavel Degterev
☐		SCRUM-44	Analyze the Metrics	DONE	SCRUM Sprint 6	Ivan Budanov
☐		SCRUM-45	Document Findings	DONE	SCRUM Sprint 6	Pavel Degterev
☐	▼	SCRUM-37	Code Clean-Up	IN PROGRESS	SCRUM Sprint 6	
☐		SCRUM-46	Refactor Based on Review	IN PROGRESS	SCRUM Sprint 6	Pavel Degterev
☐		SCRUM-47	Check Code Quality Standards	DONE	SCRUM Sprint 6	Ivan Budanov
☐		SCRUM-48	Verify Code Changes	DONE	SCRUM Sprint 6	Ivan Budanov
☐	▼	SCRUM-34	User Testing	DONE		
☐	▼	SCRUM-38	Plan Acceptance Testing	DONE	SCRUM Sprint 6	
☐		SCRUM-49	Define Acceptance Criteria	DONE	SCRUM Sprint 6	VT Viet Tran
☐		SCRUM-50	Design Acceptance Tests	DONE	SCRUM Sprint 6	SA Saed Abukar

IN PROGRESS 1

Code Clean-Up
CODE MAINTENANCE 15 APR
SCRUM-37

DONE 8

Statistical Code Review
CODE MAINTENANCE 09 APR
SCRUM-36

Plan Acceptance Testing
USER TESTING 15 APR
SCRUM-38

Create the language bundle
LOCALIZATION 24 MAR
SCRUM-20

Identification of Localization Resources
LOCALIZATION 19 MAR
SCRUM-21

Localize the User Interface
LOCALIZATION 28 MAR
SCRUM-6

Initial Sprint Planning and Sprint Review Report
LOCALIZATION 31 MAR
SCRUM-24

Sprint and Product Backlog Update
LOCALIZATION 21 MAR
SCRUM-27

Hours Contributed by Team

Tuesday	2	2	-	1
Wednesday	-	-	2	-
Thursday	-	-	-	1
Friday	3	-	-	-
Saturday	2	-	-	-
Sunday	-	-	-	1
Monday	-	-	1	1
Tuesday	1	-	-	2
Wednesday	-	1	1	2
Thursday	2	-		
Friday	2	3		
Saturday	2	-		
Sunday	-	1		
Monday	-	6	8	-

Sprint 6 hours

Viet

Pavel

Ivan

Saed

14

13

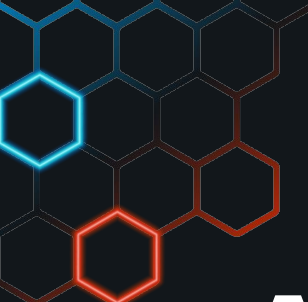
12

10



Challenges Faced

- **SpotBugs** mutable object warnings were tricky and time-consuming to resolve
 - **Acceptance Test Plan** was created partially in the dark, as some project materials or final specifications were not yet available
 - **SonarQube** (as mentioned before)
- 
- 



Ai Tools Used

- **ChatGPT / DeepSeek** = Assisted in refactoring code, writing Javadocs, and resolving static analysis issues
 - **NapkinAI** = Visuals for Presentations
- 
- 



Thanks!

